

Chlorogenic Acid from Hawthorn Berry (*Crataegus pinnatifida* Fruit) Prevents Stress Hormone-Induced Depressive Behavior, through Monoamine Oxidase B-Reactive Oxygen Species Signaling in Hippocampal Astrocytes of Mice

Dong Wook Lim, Taewon Han, Jonghoon Jung, Yuri Song, Min Young Um, Minseok Yoon, Yun Tai Kim, Seungmok Cho, In-Ho Kim, Daeseok Han, Changho Lee,* and Jaekwang Lee*

Scope: Oxidative stress has been implicated in mental disorders, including depression. Chlorogenic acid (CGA), one of the abundant phenolic compounds in herbs and fruits, has the properties of a natural antioxidant and free-radical scavenger. Therefore, we investigated the antidepressant-like effects and active mechanisms of CGA from the extract of *Crataegus pinnatifida* (CP) fruit.

Methods and results: Depression-like phenotypes were induced in mice by daily injection of stress hormone for 1–2 weeks. The brains of these animals exhibited reduced brain-derived neurotrophic factor expression and increased astrocytic hypertrophy, which are typical markers of depression in animal models. Stress hormone injection 1) upregulated monoamine oxidase B (MAOB) expression and 2) reduced spine numbers along neuronal dendrites, which indicates synaptic depression. The oral administration of CGA (30 mg kg⁻¹) or CP (300 mg kg⁻¹) prevented MAOB activation following reactive oxygen species (ROS) production and had an ameliorative effect on depressive behavioral tests (e.g., tail suspension and forced swim tests). In vitro assays performed on cultured C8-D1A cells revealed that CGA and CP inhibited MAOB activity and ROS production.

Conclusion: Our study indicates that CGA and CP extracts prevented depressive behavior and thereby have potential as natural antidepressants.

second leading cause of disability according to the most recent annual Global Burden of Disease study.^[1] Despite recent progress in the development of clinically relevant antidepressant drugs, currently available antidepressants are not totally effective and are associated with many undesirable adverse effects.^[2] Recently, alternative approaches, including the usage of traditional medicinal herbs and dietary supplements, have been of interest as novel compounds to control depression.^[3–5]

Elevation of stress hormone, such as corticosterone (CORT), resulting from the dysfunction of the hypothalamic–pituitary–adrenal (HPA) axis is one of the most prominent neurobiological findings in the pathogenesis of depression.^[6,7] Thus, increased HPA axis activity, glucocorticoid resistance, and increased plasma levels of cortisol are commonly observed in depression. In addition, high levels of glucocorticoids have been demonstrated to induce depressive-like behaviors in animal models, as indicated by decreased sucrose

consumption and significantly increased immobility time in the forced swim and tail suspension tests.^[8,9] Moreover, glucocorticoid overexposure in animals has been implicated in hippocampal dysfunction, neuronal loss,^[10] and increased expression of

1. Introduction

Depression, as the most prevalent chronic mental disorder, has an enormous impact on society. Indeed, depression is now the

Dr. D. W. Lim, T. Han, J. Jung, Y. Song, Dr. M. Y. Um, Dr. M. Yoon, Dr. Y. T. Kim, Dr. S. Cho, Dr. I.-H. Kim, Dr. D. Han, Dr. C. Lee, Dr. J. Lee
Division of Functional Food Research
Korea Food Research Institute
Jeollabuk-do 55365, Republic of Korea
E-mail: chang@kfri.re.kr; jklee@kfri.re.kr

T. Han
Department of Food and Biotechnology
Korea University
Sejong 30019, Republic of Korea
M. Y. Um
Division of Food Biotechnology
University of Science & Technology
Daejeon 34113, Republic of Korea

DOI: 10.1002/mnfr.201800029

monoamine oxidase (MAO) subtype A and B.^[11] Several studies have reported that treatment with CORT causes a significant decrease in brain-derived neurotrophic factor (BDNF) expression in the hippocampus and frontal cortex.^[12–14] It has been reported that a high concentration of glucocorticoid can induce cellular damage in PC12 cells, which possess typical neuronal features and highly express glucocorticoid receptors (GR).^[15] Various antidepressants have been demonstrated to protect PC12 cells against cytotoxicity induced by glucocorticoids.^[16] These findings suggest that a glucocorticoid-induced depression model is valid for evaluating the efficacy of potential antidepressants and exploring the mechanism of action of antidepressants.

Crataegus pinnatifida (CP) fruits, known as hawthorn berry, are distributed throughout China, Europe, and North America.^[17] *C. pinnatifida* fruits have been traditionally used as a gastric ulcer agent in traditional oriental medicine and recently as a soft drink material for diet.^[18] In Europe, hawthorn fruits, leaves, or flowers are traditionally used as astringent, antispasmodic, cardiogenic, hypotensive, and diuretic agents.^[19] Several studies on *C. pinnatifida* have reported that their various biological benefits owe to their anti-inflammatory,^[20] antioxidant,^[21] hypolipidemic,^[22] and antiobesity^[23] properties. These beneficial effects are mainly attributed to *C. pinnatifida*'s flavonoid and proanthocyanidin components, such as hyperoside, isoquercitrin, and chlorogenic acid (CGA).^[24,25] Because oxidative stress has been implicated in the pathophysiology of depression,^[26] dietary polyphenols are used as neuroprotective agents to reduce anxiety and manage depression.^[27] However, despite extensive pharmacological studies, the antidepressant effects of *C. pinnatifida* fruit or its active compounds have not been well characterized.

In this study, we explored the antidepressant efficacy and mechanisms of action of *C. pinnatifida* and its bioactive component, finding that the CP extract and CGA act as an antidepressant that reduces oxidative stress by inhibiting MAOB in hippocampal astrocytes in an animal model of stress hormone-induced depression.

2. Experimental Section

2.1. Preparation of *C. pinnatifida* Fruit Extract

Dried *C. pinnatifida* fruit was purchased from Kapdang Co. (Seoul, Korea). A dried fruit of *C. pinnatifida* (600 g) was extracted in a reflux apparatus by using 70% ethanol (6000 mL) at 80 °C for 6 h. The process was repeated once; the extracts were combined and filtered through a membrane filter (0.45 µm; Millipore, Billerica, MA, USA). After removing the solvents via rotary evaporation, the remaining extracts were freeze-dried, yielding approximately 25.1% of the dried fruit weight, w/w. The crude extract powder (100 mg) was dissolved in 50% methanol (20 mL) and filtered through a 0.45-µm filter (Sartorius Stedim, Goettingen, Germany). Subsequently, 10 µL of the solution was injected into an HPLC system (Jasco, Hachioji, Tokyo, Japan) comprising a PU-980 pump, an AS-950-10 autosampler, and an MD-2010 Plus multiwavelength detector. Chromatic separation was conducted at 30 °C on a Waters Symmetry C18 5-µm column, (4.6 × 250 mm) with gradient elution using a mobile phase composed of water, 0.1% v/v acetic acid, and acetonitrile. The analytes were

eluted from the column using a linear gradient of acetonitrile, which increased from 0% to 15% v/v within 10 min, held at 60% for an additional 40 min, and then returned to the initial gradient. The run time was set at 45 min, the flow rate was set at 1.0 mL min⁻¹, and the samples were detected at 280 nm. The concentration of CGA and epicatechin were found to be 10.8 ± 0.84 mg g⁻¹ and 18.5 ± 0.32 mg g⁻¹ using the peak area in the chromatogram and regression equation (Figure S1, Supporting Information).

2.2. Drugs and Treatment

The drugs used in the present study were CORT, dexamethasone (DEX), selegiline (SEL, a MAOB inhibitor), and CGA (Sigma Chemical Co., St. Louis, MO, USA). CORT or DEX was dissolved in 0.9% w/v saline containing 1% dimethyl sulfoxide and 1% Tween-80. SEL and CGA were dissolved in distilled water. After 1 week of adaptation, the mice were randomly clustered into six groups: 1) sham group, 2) control group, 3) 10 mg kg⁻¹ SEL-treated group, 4) 10 mg kg⁻¹, 5) 30 mg kg⁻¹ CGA-treated group, and 6) 300 mg kg⁻¹ *C. pinnatifida* extract-treated group (c.f. studies of DEX-treated mice were limited to evaluations of the effects of CP extract on depressive behavior and immunohistochemistry). Mice in the control and sample-treated groups received an intraperitoneal (i.p.) injection of CORT or DEX (40 mg kg⁻¹ of each) once daily for 7 days to induce depressive behavior, while those in the sham group received an i.p. injection of an equal volume of vehicle. The mice were treated for 7 consecutive days after CORT or DEX injection. On day 7, the mice underwent the behavioral experiments beginning 1 h after the final administration of compound. The number of mice for statistical analyses of CORT and DEX treatment was 10 per each condition. To determine relevant dose of chemical (SEL) and bioactive compound (CGA and CP extract) in the current study, we translated the doses for animals from human studies according to previous reports.^[28–30]

2.3. Animals

Male ICR mice (8 weeks old; Koatech Animal Inc., Korea) weighing 20–25 g were housed at five mice per cage under a controlled temperature (21 ± 2 °C) and a 12-h light/dark cycle (lights on at 7:00 a.m. and lights off at 7:00 p.m.). The mice were allowed at least 1 week for accommodation before the experiments. All animal protocols were approved by the Institutional Animal Care and Use Committee of the Korea Food Research Institute (KFRI-M-16038).

2.4. Behavioral Tests

2.4.1. Open-Field Test

To study spontaneous locomotor activity, mice were placed in the center of the 50 × 50 × 50 cm arena, and their behaviors were recorded and measured for 5 min with the SMART video-tracking system (SMART v3.0, Panlab SL, Barcelona, Spain). The experiment was repeated for three consecutive days at the same time each day.

2.4.2. Passive Avoidance Test

The passive avoidance task (GEMINI avoidance system, SD instruments, San Diego, CA, USA) was conducted in two compartments, one illuminated and one dark chamber (size, 24 × 20 × 20 cm). The illuminated chamber contained a light bulb, and the floor of the dark chamber compartment was composed of 2-mm steel rods. During the training trial, each mouse was placed in the illuminated compartment, and the door between the two compartments was opened. When the mice entered the dark compartment, the door closed automatically, and the mice received an electrical foot shock (0.5 mA for 3 s). On the next day, the mice were again placed in the illuminated compartment, and the time taken for them to return to the dark compartment was measured as the step-through latency (with a maximum of 300 s). If the mouse did not enter the dark compartment within 300 s, the mouse was assumed to have accomplished passive avoidance training after one training trial.

2.4.3. Forced Swim Test

This test was conducted as previously described.^[31] Briefly, mice were forced to swim for 6 min in a transparent Plexiglas cylinder (height, 13 cm and diameter, 24 cm) filled to a depth of 10 cm with water (temperature, 22–24 °C). The immobility time during the last 4 min was measured by a SMART video-tracking system (SMART v3.0, Panlab SL, Barcelona, Spain).

2.4.4. Tail Suspension Test

The TST was performed as described previously.^[32] Briefly, each mouse was suspended by tail using an adhesive tape to a hook connected to a strain gauge. The strain gauge picks up all movements of the mouse and transmits these to a central unit, which digitalizes the signals. The immobility time during the last 6 min was measured by an automated TST apparatus (BioSeb, Chaville, France).

2.5. Cell Culture

An astrocytic cell line (C8-D1A) was obtained from American Type Culture Collection (CRL-2541, ATCC, VA, USA), and was grown in Dulbecco's modified Eagle's medium (Gibco, Waltham, MA, USA), supplemented with 10% fetal bovine serum (Gibco) and 1% penicillin–streptomycin–glutamine (Gibco), using a T-75 cell culture flask (BD Falcon, Franklin Lakes, NJ, USA) in a humidified atmosphere with 5% CO₂ at 37 °C. The medium was replenished every 2–3 days. The C8-D1A cells were subcultured by trypsinization with 0.05% trypsin-EDTA (Gibco) upon reaching 80–90% confluence (split ratio, 1:5) in a T-75 flask.

2.6. Cell Viability Assay

Cell viability was analyzed by the MTT assay. Cells were seeded at a density of 2.5×10^4 cells per well in 96-well plate and incubated overnight.

On the following day, the cells were co-treated the indicated concentrations of *C. pinnatifida* extract with DEX (400 μM) for 24 h. MTT solution (Thiazolyl Blue Tetrazolium Bromide, M5655, Sigma-Aldrich) was added to each well at a final concentration of 0.5 mg mL⁻¹, and the plate was incubated at 37 °C for 3 h. The medium was subsequently discarded, and the formazan product was dissolved using DMSO (Sigma-Aldrich). Cell viability was determined by measuring the OD at 570 nm.

2.7. Reactive Oxygen Species Assay

Cellular reactive oxygen species (ROS) were estimated using 2',7'-dichlorofluorescein diacetate (DCFDA), a fluorogenic dye that measures ROS activity within the cell. An independent cell treatment was performed in 96-well plate for ROS inhibition assay, using the DCFDA Cellular ROS Detection Assay kit (ab113851, Abcam, Cambridge, UK) according to the manufacturer's instructions. Although DCF-DA fluorescence results from oxidation by potent oxidants such as superoxide, hydrogen peroxide, or nitric oxide, but it requires to be carefully interpreted the results because it could be self-amplified by redox cycling of the one-electron oxidized dye.^[33,34] Briefly, C8-D1A cells (2.5×10^4 cells per well) were seeded on a dark, clear bottom 96-well plate and incubated for 24 h before treatments. Subsequently, the cells were co-treated with CP and DEX. After 24 h, the media was removed and the cells were washed in 1 × buffer. The cells were then stained by adding 100 μL per well of the DCFDA solution (final concentration of 25 μM) and incubated for 45 min at 37 °C in the dark. DCFDA solution was removed, and the cells were washed once in 1 × buffer. Next, 100 μL per well of PBS (Gibco) was added to the wells and fluorescence was immediately measured at an excitation wavelength of 485 nm and an emission wavelength of 535 nm. The results were reported as ROS change as a percentage of control.

2.8. Monoamine Oxidase Assay

This assay is based on the detection of hydrogen peroxide in an horseradish peroxidase (HRP)-coupled reaction using 10-acetyl-3,7-dihydroxyphenoxazine (Amplex Red reagent) as a highly sensitive and stable probe for H₂O₂ that uses *p*-tyramine as a substrate. Hydrogen peroxide arising in the MAO reaction mixture was measured using an Amplex Red Monoamine Oxidase Assay kit (A12214, ThermoFisher, Waltham, MA, USA) according to manufacturer's instruction. Each reaction contained 1 mM *p*-tyramine, 1 U mL⁻¹ HRP, and 200 μM Amplex Red reagent. Reactions were incubated at room temperature (RT) for 60 min (protected from light) and fluorescence was measured with a fluorescence microplate reader at an excitation of 545 nm and an emission of 585 nm. Hydrogen peroxide produced by the MAO reaction with *p*-tyramine was detected as the fluorescence of resorufin, which was generated by the reaction of Amplex red reagent with hydrogen peroxide in the presence of HRP.

2.9. Western Blot Analysis

Mouse brains were lysed and homogenized in cold RIPA buffer (Sigma-Aldrich) with complete protease inhibitor cocktail (Roche diagnostics, Mannheim, Germany). Clear brain lysate supernatants were prepared by centrifugation at 13 000 rpm for 15 min at 4 °C and protein was quantified by the bicinchoninic acid assay (ThermoFisher). For electrophoresis and membrane transfer, 30 µg of protein from each sample was denatured in Laemmli loading buffer (Bio-Rad, CA, USA) at 100 °C for 3 min and separated on 10% SDS-PAGE at 100 V for 70 min and then transferred onto a polyvinylidene fluoride membrane (Bio-Rad, Hercules, CA, USA) at 2.5 A, 25 V for 7 min using a Trans-Blot Turbo transfer system (Bio-Rad). The membranes were blocked with 4% skim milk in TBST (20 mM Tris, pH 7.5, 500 mM sodium chloride, and 0.1% Tween 20) at RT for 40 min and were then probed with specific primary antibodies: anti-pro BDNF mouse monoclonal antibody (1:1000 dilution, sc-65514, Santa Cruz Biotechnology, CA, USA) and anti-β-actin rabbit polyclonal antibody (1:1000 dilution, #4967, Cell Signaling, MA, USA) overnight at 4 °C in 3% skim milk in TBST. The membranes were washed in TBST five times and incubated with appropriate HRP-conjugated antimouse or rabbit IgG secondary antibodies (1:2000 dilution, Vector, Burlingame, CA, USA) at RT for 2 h. Immunoreactive protein bands were detected by enhanced chemiluminescence reagent (LI-COR, Lincoln, NE, USA), and imaged using a C-DiGit Blot scanner (LI-COR). Band images were analyzed by Image Studio software (LI-COR) and normalized to β-actin levels.

2.10. Immunohistochemistry

Animals were deeply anesthetized using 2% avertin and perfused with 0.1 M PBS, followed by 4% paraformaldehyde. Brains were postfixed in 4% paraformaldehyde at 4 °C for 24 h and 30% sucrose at 4 °C for 48 h. Brains were then cut into coronal sections of 30 µm on a cryostat (Leica CM1850, Wetzlar, Germany). Sections were blocked in 0.1M PBS containing 0.3% Triton X-100 (Sigma) and 2% serum (Donkey and Goat mixture, 1:1, Abcam) for 1 h at room temperature. Primary antibody was then applied at the appropriate dilution (chicken anti-glial fibrillary acidic protein [GFAP] at 1:1000 [ab4674, Abcam], goat anti-MAOB at 1:500 [ab118865, Abcam]) and incubated overnight at 4 °C. After this, the sections were washed three times in 0.1 M PBS and incubated in secondary antibody (Alexa 488 donkey anti-chicken IgG at 1:500 [ab150173, Abcam], Alexa 555 donkey anti-goat IgG at 1:500 [ab150130, Abcam] for 2 h). After three rinses in 0.1 M PBS and the sections were mounted on polysine microscopic slide glass (Thermo Scientific) with fluoroshield with DAPI (F6057, Sigma). Images were acquired using a Nikon A1R confocal microscope and analyzed with ImageJ software.

2.11. Golgi–Cox Staining

To label dendritic branches and spines of neurons, we used the *superGolgi* Kit (003010, Bioenno Tech, LLC, Santa Ana, CA, USA). All procedures for staining were guided the protocols provided

by manufacture. Briefly, freshly removed brain tissues from mice were impregnated for 10 days and then sectioned to 60 µm thickness on a cryostat (Leica CM1850, Germany). Sectioned brain tissues were mounted on microscope slides and stain for 20 min in closed staining jar. After staining procedure, sections were dehydrated with 100% ethanol for 10 min × 4 times and then clear in xylene for 10 min × 3 times. Images were acquired using Olympus BX51 microscope and analyzed with ImageJ software (NIH, Bethesda, MD, USA).

2.12. Statistical Analyses

Statistical analyses were performed using Student's *t*-test for two sample comparisons and one-way analysis of variance, followed by Tukey's post hoc test using Prism 5 (GraphPad Software, Inc., San Diego, CA, USA) for multigroup comparisons. All data are presented as the mean ± SEM. Differences with *p* < 0.05 were considered significant.

3. Results

3.1. Induction of Depressive Behavior by Injection of Stress Hormone

To generate an animal model of depression, mice were treated with DEX (CORT analog) or CORT, the mouse analog of the human stress hormone cortisol. Behavioral tests and immunohistochemical analyses were performed to confirm depressive symptoms and pathological findings, respectively (Figure 1A). Administration of stress hormone did not have a significant effect on body weight (data not shown). There was no significant difference between the groups in the time spent in center/periphery area on the open-field test of locomotion or in step through latency on the passive avoidance test (Figure 1B,C). Stress-hormone treatment did have significant effects on depressive-like behavior on the TST and FST. Immobility time was significantly increased in the stress-hormone treated group, which is indicative of a depressive phenotype (TST: sham 118.76 ± 14.33, CORT 182.15 ± 23.94, *N* = 10 of each, **p* = 0.036; FST: sham, 106.58 ± 14.50, CORT 170.34 ± 21.11, *N* = 10 of each, **p* = 0.023). For further confirmation, we performed western blot analysis of hippocampal tissue for DEX-treated animals and observed a reduction in BDNF level (Figure 1F,G). These results indicated administration of stress-hormone for 2 weeks successfully manifested a mouse model of depression.

3.2. Stress Hormone Induced Astrocytic Hypertrophy is Prevented by CP Treatment

It was previously reported that astrocytes are transformed to their reactive form (hypertrophy) by mild and/or chronic stress, making this process a neural marker of depression.^[35] Therefore, we examined whether DEX injection affected astrocytic hypertrophy in the mouse brain through immunohistochemical analysis. Immunohistochemical data revealed that hippocampal

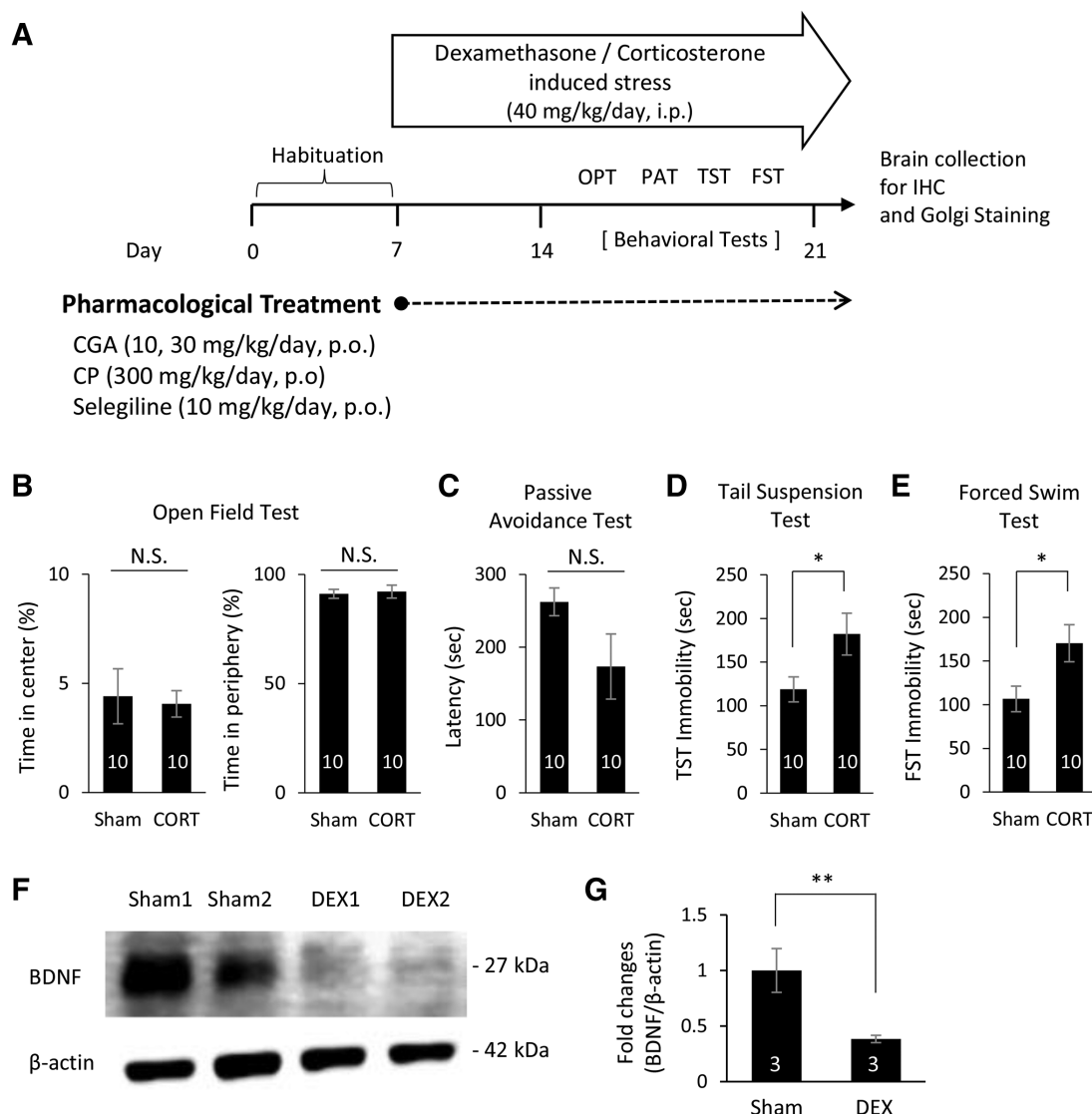


Figure 1. A) Experimental scheme for generating the CORT-induced depression-like behavior animal model. B) Effect of repeated CORT 40 mg kg⁻¹ injection on locomotor activity, C) passive avoidance latency time, and immobility time on the D,E) TST and FST, respectively ($N = 10$ per group). Image showing hippocampal BDNF levels as determined by F) western blotting with G) quantification ($N = 3$ per group). Results are presented as mean $\pm$ SEM. $^{**}p < 0.01$ versus the sham group. CORT, corticosterone; DEX, dexamethasone; BDNF, brain-derived neurotrophic factor; OPT, open-field test; PAT, passive avoidance test; TST, tail suspension test; FST, forced swim test.

astrocytes in the DEX group increased the expression of GFAP and GR compared with the sham group (Figure 2A,D). Sholl analysis was performed to determine the extent of DEX-induced astrocytic hypertrophy. The number of intersections of astrocytic branches also increased in the DEX-treated group (no. of intersections: sham 12.0 ± 0.93 , $N = 10$, DEX 16.5 ± 1.22 , $N = 9$, DEX + CP 11.6 ± 0.99 , $N = 11$, $^{**}p = 0.0079$, Sham vs DEX; $P = 0.0047$, DEX vs DEX + CP; Figure 2B,C). These increases of GFAP, GR, and branch number were prevented by co-treatment with CP extract (Pixel Density, control of sham [%]: sham = 100, DEX = 341.07, DEX + CP = 71.96; Figure 2A–E). Moreover, we found that MAOB expression was increased in hippocampal astrocytes of DEX-treated mice. This increase in the expression of MAOB was reduced by co-treatment with CP. Altogether with the behav-

ioral findings, these data indicate that increased stress hormone induced astrocytic hypertrophy and this correlated with induction of depressive-like symptoms. Moreover, CP treatment was found to have an inhibitory effect on astrocytic hypertrophy and MAOB expression in the stress hormone-induced depression model.

3.3. CGA and CP Have Antioxidative Effect via Inhibition of Monoamine Oxidase B

To investigate the mechanism of glial cell-mediated depressive phenotypes, we performed in vitro assays on the C8-D1A cell line (astrocyte type I cell line; Figure 3A). To evaluate a possible

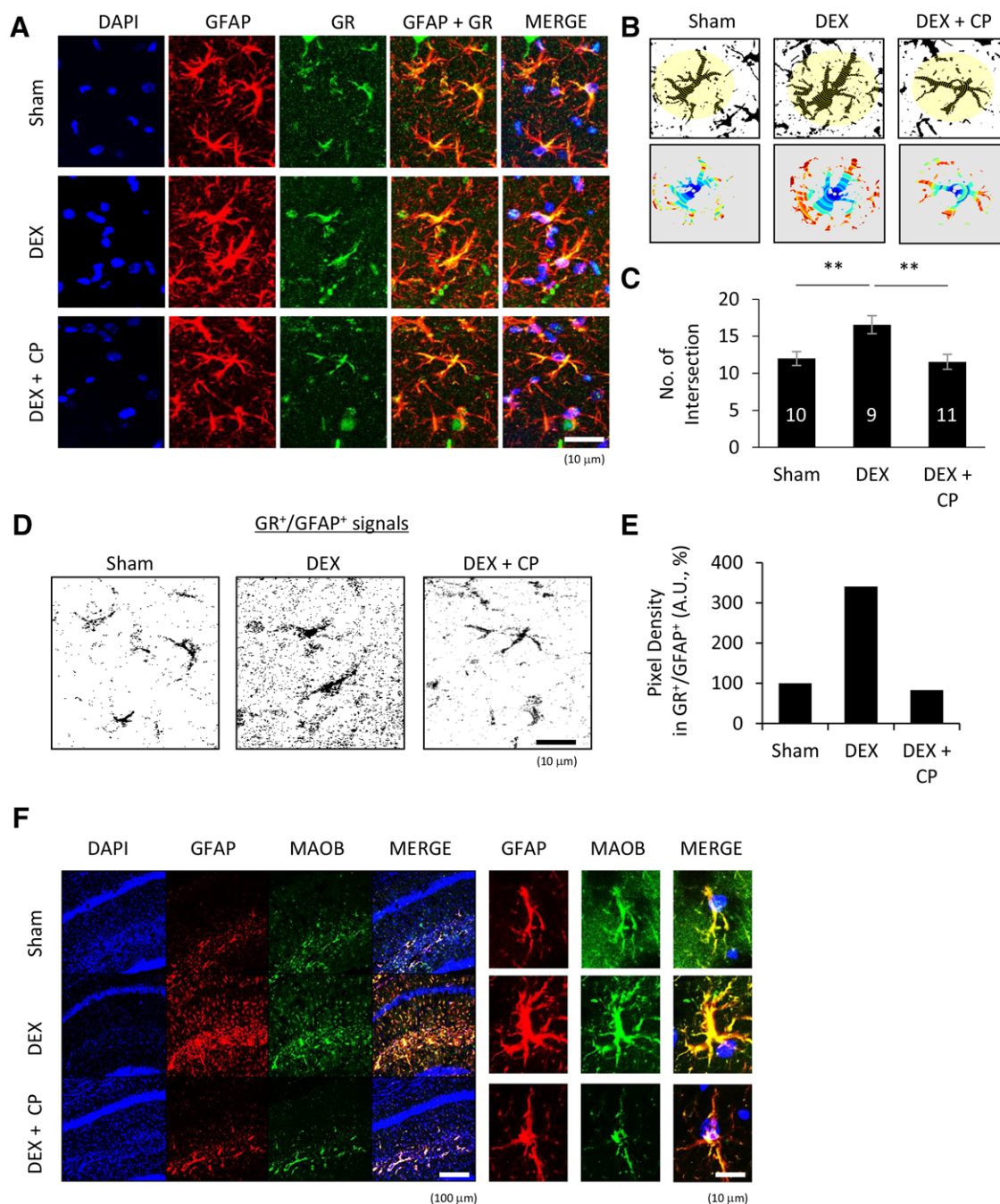


Figure 2. Inhibitory effects of CP extract on DEX-induced astrocytic hypertrophy in mouse brain using immunohistochemical analysis. Micrographs showing DEX-induced increases in A) GFAP expression, B,C) astrocytic branching, D,E) GR levels, and F) MAOB expression were prevented by co-treatment with CP extract. Results are presented as mean $\pm$ SEM. $^{**}p < 0.01$ versus the DEX group. CP, *C. pinnatifida* fruit; DEX, dexamethasone; DAPI, diaminodiphenylindole; GFAP, glial fibrillary acidic protein; GR, glucocorticoid receptors; MAOB, monoamine oxidase subtype B.

single molecule from CP extract has an effect on depressive phenotypes, we tested CGA, a known bioactive component of CP (Figure S1 Supporting Information). First, we assessed cell viability in a DEX-induced stress condition. C8-D1A cells incubated with DEX exhibited reduced the viability by approximately 30% compared with control, and this reduction was prevented by co-treatment with CGA or CP (cell viability [%]: CON, 100 $\pm$

2.89; DEX, 73.45 $\pm$ 1.64, CGA 92.38 $\pm$ 3.76; 10 CGA 90.01 $\pm$ 2.87; 100 CP 92.77 $\pm$ 0.87; 200 CP 91.51 $\pm$ 1.74, $^{***}p = 0.0009$ (CON vs DEX); Figure 3B).

It was previously shown that DEX treatment induced an increase in the expression of MAOB in vivo and that the activation of MAO enzyme increased reactive oxygen stress (ROS) generation in both in vitro and in vivo models.^[36] Therefore, we next

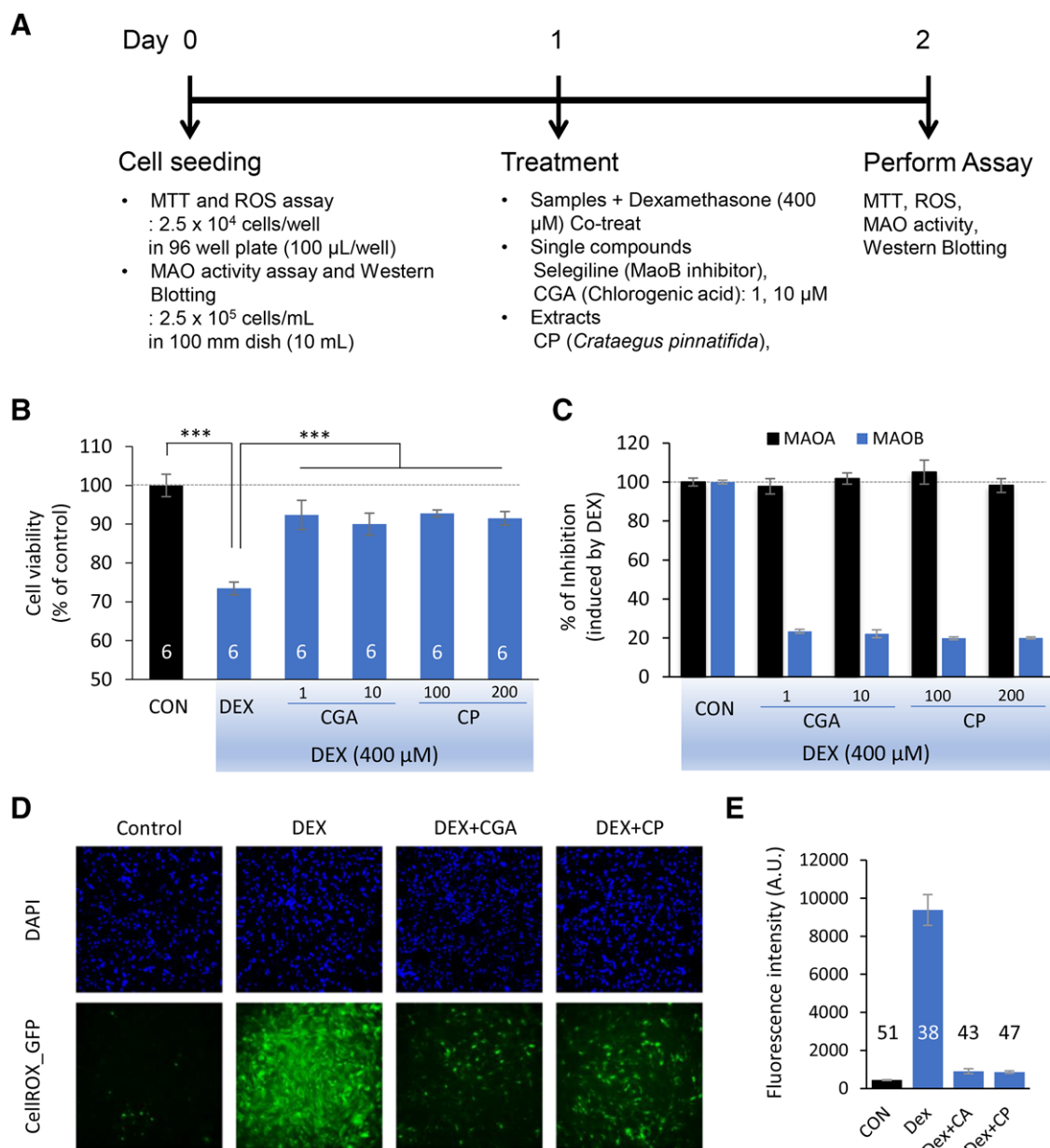


Figure 3. A) Experimental scheme of in vitro assays of DEX-induced reduction in cell viability. B) Reduction in cell viability was prevented by co-treatment with CGA and CP extract in DEX-treated C8-D1A cells. C) MAOB activity was significantly decreased by co-treatment of DEX-treated C8-D1A cells with CGA or CP extract. D) Quantitative imaging analysis of cellular ROS detected by CellROX green reagent applied to DEX-treated C8-D1A cells. E) The increase of fluorescence in the DEX-treated condition was almost completely inhibited by co-treatment of CGA and CP. Results are presented as mean $\pm$ SEM. *** $p < 0.001$ versus the DEX group. CP, *C. pinnatifida* fruit; CGA, chlorogenic acid; SEL, selegiline; CON, control; DAPI, diamidinophenylindole; MAO, monoamine oxidase B; ROS, reactive oxygen species; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide.

tested the effect of CGA and CP on MAO activity in C8-D1A cells. There are two subtypes of MAO (MAOA and MAOB) in the brain, although MAOB is known as the major form in astrocytes.^[37] We thus first determined which subtype of MAO enzyme is important for ROS production in C8-D1A cells by performing MAO enzyme activity assays. We found that MAOB activity was significantly decreased upon treatment with CGA (1 and 10 μ M) or CP (100 and 200 μ g g^{-1}) in DEX-treated C8-D1A cells (inhibition [%]: CON, 100.00 $\pm$ 0.92; 1 CGA, 23.36 $\pm$ 1.00; 10 CGA, 22.13 $\pm$ 2.02;

100 CP, 19.84 $\pm$ 0.73; 200 CP, 20.02 $\pm$ 0.54, $N = 3$, *** $p < 0.001$). However, the activity of MAOA was not different than control (Figure 3C). This result indicated that CGA and CP selectively inhibited the activity of MAOB, rather than MAOA in C8-D1A cells. Next, we performed ROS signal assays using fluorescence detecting tools. GFP fluorescence signal was markedly increased in cultured C8-D1A cells incubated with DEX compared with untreated cells. The increase of fluorescence in the DEX-treated condition was almost completely inhibited by co-treatment with CGA, CP,

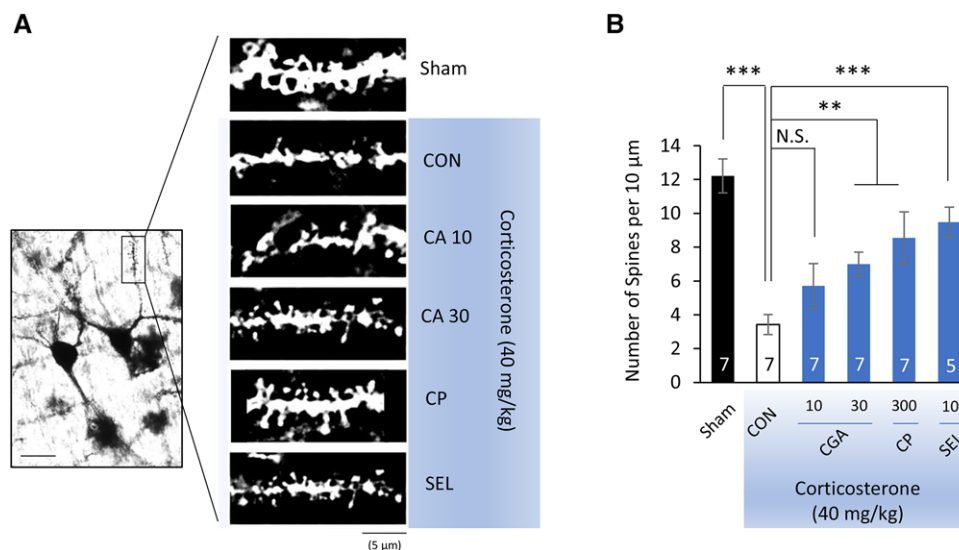


Figure 4. Effects of CGA and CP extract on CORT-induced dendritic spine density decline observed by Golgi–Cox staining. A,B) Image showing that co-treatment with CP extract and SEL prevented CORT-induced declines in spine numbers in hippocampal dendrites. Results are presented as mean $\pm$ SEM. *** p < 0.001 and ** p < 0.01 versus the CON group. CP, *C. pinnatifida* fruit extract; CGA, chlorogenic acid; SEL, selegiline; CON, control.

or SEL, a selective MAOB inhibitor (Figure 3D,E). These results indicated that CGA and CP have a protective effect on ROS stress through MAOB inhibition in C8-D1A cells (Figure 3D).

3.4. CGA and CP Prevented Loss of Dendritic Spine Numbers in Hippocampal Neurons

Oxidative stress may impair the plasticity of dendritic spines, which may consequently cause synaptic weakening. Therefore, we investigated whether stress-hormone induced oxidative stress alters spine density. After behavioral testing, mouse tissue was harvested and we performed Golgi–Cox staining of whole brains from conditioned animals to examine the dendritic spines of hippocampal neurons. Stress hormone (CORT)-injected mice displayed significantly reduced dendritic spine density in hippocampal neurons compared with the sham group (spine numbers: sham, 12.21 ± 1.01 , $N = 7$; with CORT, CON, 3.43 ± 0.59 , $N = 7$). However, co-treatment with CP prevented these declines in spine numbers along hippocampal dendrites (Figure 4A). CGA and SEL also showed similar effects on the number of spines of hippocampal neurons to CP extract (spine numbers; with CORT, sham, 12.21 ± 1.01 , $N = 7$; CON, 3.43 ± 0.59 , $N = 7$; 10 CGA, 5.71 ± 1.32 , $N = 7$; 30 CGA, 7.0 ± 0.71 ; 300 CP, 8.55 ± 1.54 , $N = 7$; SEL, 10.30 ± 1.30 , $N = 5$) (Figure 4B). These results indicate that MAOB activation leads to loss of hippocampal dendritic spine density downstream of stress hormone and that CP may have an inhibitory effect on the MAOB signaling pathway.

3.5. CGA and CP Have an Antidepressive Effect In Vivo

Next, we tested whether CGA and CP have antidepressive effects in vivo. The CORT-injected group showed significantly increased immobility time when we performed tail suspension test (TST)

and forced swim test (FST), which are established methods to determine depressive-like behavior in rodents. During these tests, the increased immobility time associated with CORT treatment was significantly reduced by the oral administration of CGA or CP, reducing immobility by up to 53.1%, and 46.2% at doses of 30 mg kg⁻¹ (for CGA) and 300 mg kg⁻¹ (for CP). The reduction of immobility time in both tests was also observed when SEL (10 mg kg⁻¹) was administered (immobility time for TST (s): sham, 114.73 ± 13.40 ; with CORT, CON, 186.40 ± 15.32 ; 10 CGA, 153.06 ± 21.23 ; 30 CGA, 87.40 ± 13.61 ; 300 CP, 100.37 ± 11.27 ; SEL, 80.92 ± 18.15 , $N = 10$ of each; p -values, sham versus CON ($p = 0.0356$); CON versus 10 CGA ($p = 0.1689$), 30 CGA ($p = 0.0035$), 300 CP ($p = 0.0082$), and SEL ($p = 0.0039$); immobility time (s) for FST: sham, 106.58 ± 14.50 ; with CORT, CON, 170.34 ± 21.11 ; 10 CGA, 122.90 ± 22.54 ; 30 CGA, 99.91 ± 17.45 ; 300 CP, 112.48 ± 15.76 ; SEL, 74.70 ± 17.39 , $N = 10$ of each; p -values, sham versus CON ($p = 0.0228$); CON versus 10 CGA ($p = 0.1419$), 30 CGA ($p = 0.0192$), 300 CP ($p = 0.0414$), and SEL ($p = 0.0026$) (Figure 5). These results strongly indicate that CGA and CP have antidepressive effects in vivo as well as in vitro through inhibition of MAOB.

4. Discussion

To our knowledge, the present study is the first investigation for the antidepressant-like effects of CGA from *C. pinnatifida* fruit extracts in mice showing depressive-like behaviors induced by stress hormone, a well-validated animal model for human depression. CGA and CP extracts had an antidepressive effect on behaviors as evidenced by a significant decrease in immobility times on the TST and FST, without affecting locomotor activity. Moreover, CGA significantly alleviated CORT-induced astrocytic hypertrophy and reduced spine numbers along dendrites of hippocampal neurons. As a potential mechanism, treatment with

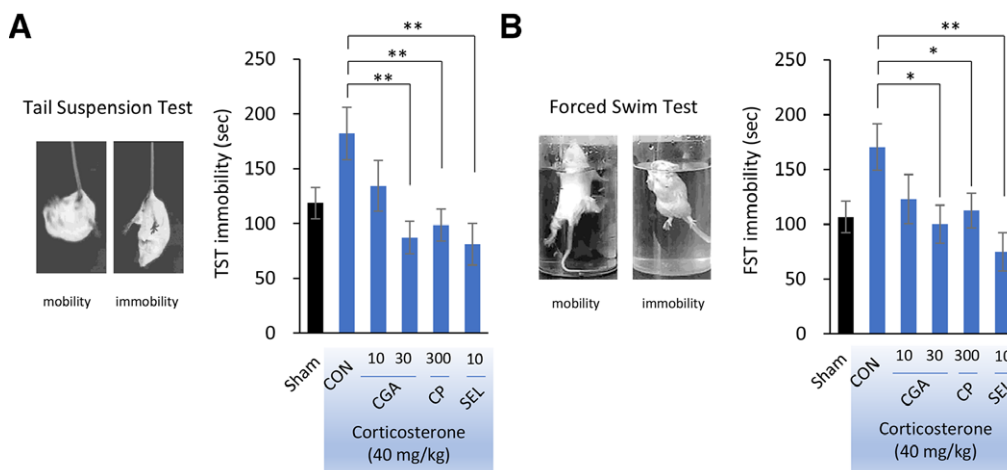


Figure 5. Effects of CGA and CP extract on CORT-induced depression-like behavior in response to the TST and FST. A,B) Increased immobility time in response to CORT injection was significantly reduced by oral administration of CGA (30 mg kg⁻¹) or CP (300 mg kg⁻¹) (N = 10 per group). Results are presented as mean ± SEM. ***p* < 0.01 and **p* < 0.05 versus the CON group. CP, *C. pinnatifida* fruit; CGA, chlorogenic acid; SEL, selegiline; CON, control.

CGA and CP extracts evidenced inhibitory effects on MAOB activity and ROS production in stress hormone-treated C8-D1A cells. These results suggest that CGA from CP extracts could ameliorate CORT-induced depressive behavior and alteration of neuronal morphology through protection against glial activation and oxidative stress.

According to previous reports, continuous elevation of CORT levels resulting from dysfunction of the HPA axis is one of the most consistent neurobiological findings in the pathogenesis of depression. GR mediates the direct effects of glucocorticoids, which are released in response to stress and regulate the HPA axis via a negative feedback mechanism.^[38,39] Previous clinical studies have reported that depressed subjects exhibited downregulation of GR expression, which subsequently led to an increase in endogenous levels of glucocorticoids.^[40] Thus, increased HPA axis activity and glucocorticoid resistance, as well as hypercortisolemia, are commonly observed in depression.^[41]

Animal models for depression play an important role in the screening and evaluation of antidepressants. The TST and FST are effective screening tools with good reliability and predictive validity^[42]; the state of immobility in these tests is reported to mimic the symptoms of depression in humans and can be reversed by treatment with antidepressant drugs.^[43] Anhedonia, a core symptom of major depression in humans, is modeled by inducing a decrease in response to rewards, as reflected by reduced preference for sweetened solutions.^[44] A sucrose preference test has been used as an indicator of anhedonia-like behavior to measure the level of anhedonia in an anhedonia animal model.^[45] Injection of a high level of glucocorticoid has been demonstrated to induce depressive-like behaviors in animal models, as indicated by the significant increase in immobility time on the TST and FST and decrease of preference for sucrose consumption. Michihiko et al. reported that antidepressants or a GR antagonist, mifepristone, potently reduced depressive behavior in the acute or chronic CORT treatment-induced animal model.^[46] Their findings implied that TST and FST are valid tools for

evaluating the efficacy of potential antidepressants and exploring their mechanisms of action.^[47] Our study clearly showed that acute treatment of CGA and CP extract significantly decreased immobility time of stress hormone-induced depressive mice on the TST and FST. These results suggest that CGA and CP extract exhibit significant antidepressant-like effects on the stress hormone-induced depressive animal model. However, to verify that CGA and CP extract act as antidepressants and to determine the potential role of CP bioactive compound(s) on anhedonia, further investigations employing several experimental approaches are required, including sucrose preference or intracranial self-stimulation.

Glial cell MAO activity has been implicated as a contributor to oxidative neuronal damage associated with various neurodegenerative diseases such as Alzheimer disease and Parkinson's disease.^[48,49] Therefore, the attenuation of MAO activity may provide protection against oxidative neurodegeneration. In the present study, we provide substantial evidence that CP extract and CGA have selective inhibitory effects on MAOB activity. An elevation in astrocyte MAOB activity in proximate association with degenerating neurons in human diseases of the CNS was previously reported.^[50,51] Spine loss, which is considered a potential underlying mechanism of depressive symptoms, can alter neuroplasticity and levels of neurotransmitters.^[52] In the current study, loss of spine numbers in stress hormone-injected depressive animals was markedly recovered by the inhibition of MAOB activity and by CP extract and CGA treatment. This protective effect might possibly be because of CP extract and CGA-induced reductions in MAOB activity in astrocytes. Notably, food constituents that reduce MAOB activity in astrocyte cultures have previously been reported.^[53,54]

In conclusion, previous findings combined with the current evidence reported in this study strongly support the idea that digestive consumption of particular foods and food components are able to regulate astrocytic MAOB activity and protect against clinical depression.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

D.W.L. and T.H. contributed equally to this work. D.W.L., T.H., J.L., and C.L. designed the study. J.J., M.Y.U., Y.T.K., I.H.K., D.H., and S.C. conducted experiments and analyzed data. D.W.L., J.L., and C.L. contributed to manuscript drafting. All authors reviewed the manuscript and approved the final version. This study was supported by the Main Research Program (E0164501-02 and E0133116-05) of the Korea Food Research Institute funded by the Ministry of Science and ICT and by a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare, Republic of Korea (grant number: HI17C0952).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

antidepressant, brain-derived neurotrophic factor, chlorogenic acid, *Crataegus pinnatifida*, oxidative stress

Received: January 10, 2018

Revised: April 11, 2018

Published online: July 11, 2018

- [1] S. Moussavi, S. Chatterji, E. Verdes, A. Tandon, V. Patel, B. Ustun, *Lancet* **2007**, 370, 851.
- [2] R. S. Duman, G. R. Heninger, E. J. Nestler, *Arch. Gen. Psychiat.* **1997**, 54, 597.
- [3] L. Butler, K. Pilkington, *Evid-Based Compl. Alt. Med.* **2013**, 2013, 739716.
- [4] S. E. Lakhan, K. F. Vieira, *Nutr. J.* **2010**, 9, 42.
- [5] J. Sarris, A. Panossian, I. Schweitzer, C. Stough, A. Scholey, *Eur. Neuropsychopharm.* **2011**, 21, 841.
- [6] K. Twardowska, J. Rybakowski, *Psychiatr. Pol.* **1996**, 30, 741.
- [7] M. Aihara, I. Ida, N. Yuuki, A. Oshima, H. Kumano, K. Takahashi, M. Fukuda, N. Oriuchi, K. Endo, H. Matsuda, M. Mikuni, *Psychiatr. Res.* **2007**, 155, 245.
- [8] A. Gregus, A. J. Wintink, A. C. Davis, L. E. Kalynchuk, *Behav. Brain Res.* **2005**, 156, 105.
- [9] B. Lee, I. Shim, H. J. Lee, Y. Yang, D. H. Hahm, *Neurosci. Lett.* **2009**, 453, 151.
- [10] P. J. Lucassen, M. B. Muller, F. Holsboer, J. Bauer, A. Holtrop, J. Wouda, W. J. Hoogendijk, E. R. De Kloet, D. F. Swaab, *Am. J. Pathol.* **2001**, 158, 453.
- [11] T. A. Slotkin, F. J. Seidler, J. C. Ritchie, *Brain. Res. Bull.* **1998**, 47, 345.
- [12] Y. Dwivedi, H. S. Rizavi, G. N. Pandey, *Neuroscience* **2006**, 139, 1017.
- [13] F. Karege, G. Perret, G. Bondolfi, M. Schwald, G. Bertschy, J. M. Aubry, *Psychiatr. Res.* **2002**, 109, 143.
- [14] M. J. Schaaf, E. R. De Kloet, E. Vreugdenhil, *Stress* **2000**, 3, 201.
- [15] Y. F. Li, Z. H. Gong, M. Yang, Y. M. Zhao, Z. P. Luo, *Life Sci.* **2003**, 72, 933.
- [16] Y. F. Li, Y. Q. Liu, W. C. Huang, Z. P. Luo, *Acta Pharmacol. Sin.* **2003**, 24, 996.
- [17] H. H. Fong, J. L. Bauman, *J. Cardiovasc. Nurs.* **2002**, 16, 1.
- [18] C. Y. Chu, M. J. Lee, C. L. Liao, W. L. Lin, Y. F. Yin, T. H. Tseng, *J. Agr. Food Chem.* **2003**, 51, 7583.
- [19] Q. Chang, Z. Zuo, F. Harrison, M. S. Chow, *J. Clin. Pharmacol.* **2002**, 42, 605.
- [20] E. S. Kao, C. J. Wang, W. L. Lin, Y. F. Yin, C. P. Wang, T. H. Tseng, *J. Agr. Food Chem.* **2005**, 53, 430.
- [21] C. Li, H. J. Son, C. Huang, S. K. Lee, J. Lohakare, M. H. Wang, *Food Sci. Biotechnol.* **2010**, 19, 769.
- [22] C. Y. Kwok, C. N. Y. Wong, M. Y. C. Yau, P. H. F. Yu, A. L. S. Au, C. C. W. Poon, S. W. Seto, T. Y. Lam, Y. W. Kwan, S. W. Chan, *J. Funct. Foods* **2010**, 2, 179.
- [23] D. H. Kuo, C. H. Yeh, P. C. Shieh, K. C. Cheng, F. A. Chen, J. T. Cheng, *J. Ethnopharmacol.* **2009**, 124, 544.
- [24] P. Z. Liu, B. R. Yang, H. Kallio, *Food Chem.* **2010**, 121, 1188.
- [25] T. Jurikova, J. Sochor, O. Rop, J. Mlcek, S. Balla, L. Szekeres, V. Adam, R. Kizek, *Molecules* **2012**, 17, 14490.
- [26] G. Lucca, C. M. Comim, S. S. Valvassori, G. Z. Reus, F. Vuolo, F. Petronilho, E. C. Gavioli, F. Dal Pizzol, J. Quevedo, *J. Psychiatr. Res.* **2009**, 43, 864.
- [27] G. P. Dias, N. Cavegn, A. Nix, M. C. D. Bevilacqua, D. Stangl, M. S. A. Zainuddin, A. E. Nardi, P. F. Gardino, S. Thuret, *Oxid. Med. Cell. Longev.* **2012**, 2012, 541971.
- [28] S. Reagan-Shaw, M. Nihal, N. Ahmad, *FASEB J.* **2008**, 22, 659.
- [29] J. D. Amsterdam, J. Shults, *J. Affect. Disorders* **2005**, 89, 183.
- [30] E. Koch, F. A. Malek, *Planta Med.* **2011**, 77, 1123.
- [31] B. Petit-Demouliere, F. Chenu, M. Bourin, *Psychopharmacology* **2005**, 177, 245.
- [32] J. F. Cryan, C. Mombereau, A. Vassout, *Neurosci. Biobehav. R.* **2005**, 29, 571.
- [33] H. J. Forman, O. Augusto, R. Brigelius-Flohe, P. A. Dennerly, B. Kalyanaraman, H. Ischiropoulos, G. E. Mann, R. Radi, L. J. Roberts, J. Vina, K. J. A. Davies, *Free Radical Bio. Med.* **2015**, 78, 233.
- [34] B. Kalyanaraman, V. Darley-Usmar, K. J. A. Davies, P. A. Dennerly, H. J. Forman, M. B. Grisham, G. E. Mann, K. Moore, L. J. Roberts, H. Ischiropoulos, *Free Radical Bio. Med.* **2012**, 52, 1.
- [35] M. Banasr, G. M. I. Chowdhury, R. Terwilliger, S. S. Newton, R. S. Duman, K. L. Behar, G. Sanacora, *Mol. Psychiatr.* **2010**, 15, 501.
- [36] J. K. Mallajosyula, D. Kaur, S. J. Chinta, S. Rajagopalan, A. Rane, D. G. Nicholls, D. A. Di Monte, H. Macarthur, J. K. Andersen, *PLoS One* **2008**, 3.
- [37] M. B. Youdim, P. Riederer, *J. Neural. Transm. Gen. Sect.* **1993**, 91, 181.
- [38] E. R. De Kloet, J. M. Reul, *Psychoneuroendocrino.* **1987**, 12, 83.
- [39] M. J. Webster, M. B. Knable, J. O'grady, J. Orthmann, C. S. Weickert, *Mol. Psychiatr.* **2002**, 7, 985.
- [40] C. M. Pariante, A. H. Miller, *Biol. Psychiatr.* **2001**, 49, 391.
- [41] O. M. Wolkowitz, V. I. Reus, T. Chan, F. Manfredi, W. Raum, R. Johnson, J. Canick, *Bio. Psychiatr.* **1999**, 45, 1070.
- [42] B. Petit-Demouliere, F. Chenu, M. Bourin, *Psychopharmacology (Berl)* **2005**, 177, 245.
- [43] C. E. Renard, E. Dailly, D. J. David, M. Hascoet, M. Bourin, *Fundam. Clin. Pharmacol.* **2003**, 17, 449.
- [44] A. Bergstrom, M. N. Jayatissa, A. Mork, O. Wiborg, *Brain Res.* **2008**, 1196, 41.
- [45] R. Rygula, N. Abumaria, G. Flugge, E. Fuchs, E. Ruther, U. Havemann-Reinecke, *Behav. Brain Res.* **2005**, 162, 127.
- [46] M. Iijima, A. Ito, S. Kurosu, S. Chaki, *Brain Res.* **2010**, 1359, 75.
- [47] U. Skupio, M. Tertilt, M. Sikora, S. Golda, A. Wawrzczak-Bargiela, R. Przewlocki, *Neuroscience* **2015**, 286, 141.
- [48] E. Dossi, F. Vasile, N. Rouach, *Brain. Res. Bull.* **2018**, 136, 139.
- [49] E. A. Mazzio, N. Harris, K. F. Soliman, *Planta Med.* **1998**, 64, 603.

- [50] J. Saura, J. M. Luque, A. M. Cesura, M. Da Prada, V. Chan-Palay, G. Huber, J. Löffler, J. G. Richards, *Neuroscience* **1994**, 62, 15.
- [51] S. Jo, O. Yarishkin, Y.J. Hwang, Y.E. Chun, M. Park, D.H. Woo, J.Y. Bae, T. Kim, J. Lee, H. Chun, H.J. Park, D.Y. Lee, J. Hong, S.J. Oh, H. Lee, B.E. Yoon, Y. Kim, Y. Jeong, I. Shim, Y.C. Bae, J. Cho, N.W. Kowall, H. Ryu, E. Hwang, D. Kim, C.J. Lee, *Nat. Med.* **2014**, 20, 886.
- [52] E. S. Wohleb, T. Franklin, M. Iwata, R. S. Duman, *Nat. Rev. Neurosci.* **2016**, 17, 497.
- [53] H. T. Olsen, G. I. Stafford, J. Van Staden, S. B. Christensen, A. K. Jager, *J. Ethnopharmacol.* **2008**, 117, 500.
- [54] J. S. Liu, W. Qiao, Y. Yang, L. Y. Ren, Y. Sun, S. X. Wang, *J. Ethnopharmacol.* **2012**, 141, 257.